

ECE 2049 Syllabus

ECE2049: Embedded Computing in Engineering Design D Term 2023

Instructor: Fatemeh (Saba) Ganji, fganji@wpi.edu

Lectures: MT-RF, 10:00 – 10:50 AM, AK233

Office Hours: T, 11:00 a.m.-12:00 p.m. in AK204

TAs:

Nagasai A. Kodumuru (nkodumuru@wpi.edu)
Jonathan Lopez (jrlopez@wpi.edu)
Arnav Sacheti (asacheti@wpi.edu)
Varun Walimbe (vawalimbe@wpi.edu)
Xiao Zhang (xzhang25@wpi.edu)

Timetable

ECE 2049																D23															
	6:00 a.m.	6:30 a.m.	7:00 a.m.	7:30 a.m.	8:00 a.m.	8:30 a.m.	9:00 a.m.	9:30 a.m.	10:00 a.m.	10:30 a.m.	11:00 a.m.	11:30 a.m.	12:00 p.m.	12:30 p.m.	1:00 p.m.	1:30 p.m.	2:00 p.m.	2:30 p.m.	3:00 p.m.	3:30 p.m.	4:00 p.m.	4:30 p.m.	5:00 p.m.	5:30 p.m.	6:00 p.m.	6:30 p.m.	7:00 p.m.	7:30 p.m.	8:00 p.m.		
Mon									Lecture									Xiao Zhang													
Tue									Lecture		Jonathan Lopez									Nagasai Kodumuru											
Wed																															
Thu									Lecture		Jonathan Lopez									Nagasai Kodumuru											
Fri									Lecture								Varun Walimbe										Arnav Sacheti				

* If you missed the lab session on Wednesday, you may go to any of the TA office hours

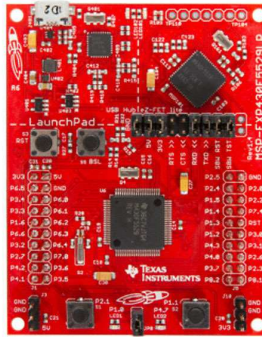
Help sessions: **Atwater Kent 317A ECE Lab**

Pre-requisite: This course is about programming embedded systems using C language. Therefore, having a good knowledge of programming in C is required. We suggest that students take CS 2301 (or its equivalent) before this course.

ECE 2049 Syllabus

Required Material: (We will provide it)

1. Texas Instruments MSP430F5529 Launchpad, *available online from TI, Mouser, Digi-key.*
 - <https://www.mouser.com/ProductDetail/Texas-Instruments/MSP-EXP430F5529LP?qs=sGAEpiMZZMvFkM%2FHBrKaDY%2FSkQ9SDcs5>
 - <https://www.digikey.com/product-detail/en/texas-instruments/MSP-EXP430F5529LP/296-36506-ND/4311683>
 - <http://www.ti.com/tool/MSP-EXP430F5529LP#buy>



Reference Texts: (not required)

"MSP 430 Microcontroller Basics" by John H. Davies.

A good C reference textbook, e.g.

"C How to Program (5-8th Eds.)" by Deitel & Deitel, Prentice-Hall, ISBN-13:9780132404167

Canvas: Course will use Canvas.wpi.edu to post lecture notes, homework, labs, and grades.

Note on Academic Integrity: Note that our class does/will not tolerate any type of academic dishonesty, including plagiarism (e.g., using other original academic material of another without providing proper citation or attribution), cheating (i.e., using or attempting to use **unauthorized materials**, information, or study aids in any academic exercise), facilitation, using ChatGPT (or another AI based software) to generate text or code for assignments etc.¹ If we suspect academic dishonesty, we will follow the procedures outlined in the [Faculty Guide to Academic Integrity at WPI](#).

Grading: Your grades will be computed as follows. Full adherence to WPI's and the department's academic honesty policies is expected.

Homework:	10%
Laboratory:	30%
Exam1:	20%
Exam2:	20%
Exam3:	20%

¹ <https://www.wpi.edu/about/policies/academic-integrity/dishonesty>

ECE 2049 Syllabus

Homework: You are encouraged to discuss homework assignments with your classmates and the TAs. However, each student must hand in their own implementation of the homework. Copying of homework will result in a grade of zero for all parties involved. **No late homework will be accepted after 11:59pm of the due dates.**

The homework assignments need to be submitted to Canvas (**use an app to take a scanned copy, i.e., create a pdf**). **Just photos will not be accepted!**

Labs: All labs will be completed in groups of two. You can post announcements on canvas if you are looking for a group member. A single lab report is satisfactory for each group. You are encouraged to discuss with the TAs or your classmates, but **the code and report must be your own solution to the lab**. Duplication of code and/or data will result in a grade of zero for all parties involved. **Lab reports are due by the following Wednesday 11:59pm. Late Labs are accepted. Each week %10 of the maximum point will be deducted. (For instance; if lab is 150 points, 15 points will be deducted). Some Labs may include bonus points if you complete bonus assignments. Also, some of the labs have early sign-off bonus (%5). However, if you are taking the early sign-off, the Lab should be complete.**

The Labs need to be submitted to Canvas (**use an app to take a scanned copy, i.e., create a pdf**). **Just photos will not be accepted!**

Exams: All exams are strictly individual efforts. Exams will be open-book and open for a window of time, from 8 a.m. to 11:59 p.m. There is no class on exam days. Exams are posted on Canvas at 8 a.m. on exam days. When you are done with the exam, you can submit your sheet to Canvas, similar to a homework assignment. If you cannot take the exam due to emergency, please contact us for make up exam.

Letter grades: $A \geq 90\%$, $75\% \leq B < 90\%$, $60\% \leq C < 75\%$, NR if less than 60%.

Submitting Handwritten Work/Exam:

Do not submit simple photos of your handwritten work to Canvas! All assignments submitted for grading must be in **PDF** format, except for source code (ZIP files).

It is greatly preferred that you use a **flatbed scanner** to scan your paper documents into 300 dpi black-and-white (color OK too) PDF files. If you do not have a scanner, use a **camera scanner app** on your phone. Adobe Scan, CamScanner and Office Lens are good apps in this regard. You may sign in using your WPI credentials. There are many other alternatives.

Please submit only **one** PDF per assignment, not each page in a separate PDF. The graders will annotate your PDF directly on Canvas.

About Class Format:

We will do in person lectures and labs. You need the custom boards that will be handed to you to complete the labs (you will receive it in Lab0), you may work from your dorm room or in the lab. It is expected that you plan this with your lab-mate.

Students with Special Needs:

Students with approved academic accommodations should plan to submit their accommodation letters through the Office of Accessibility Services Student Portal. Should you have any questions about how accommodations can be implemented in this particular course, please contact me as soon as possible. Students who are not currently registered with the Office of Accessibility Services (OAS) but who would like to find out more information about requesting

ECE 2049 Syllabus

accommodations, documentation guidelines, and what the accommodated interactive process entails should plan to contact OAS either by email AccessibilityServices@wpi.edu, by phone (508) 831-4908, or by stopping by the office on the 5th floor of Unity Hall.

ECE 2049 Syllabus

Tentative Class Schedule:

Monday	Tuesday	Wednesday	Thursday	Friday
13	14	15 No lab this week	16	17
20	21	22 Hw1-due Lab0	23	24
27 Recorded-Lecture Hw2-due	28 Wellness Day	29 Lab1 Lab0 report due	30 Recorded-Lecture Review for Exam1	31 Exam1
3	4	5 Lab2 Lab1 report due	6	7 No-Class
10 Hw3-due	11	12 Lab2	13	14
17 No-Class	18	19 Lab3 Lab2 report due	20 Hw4-due	21 No-Class
24 Review for Exam2	25 Exam2	26 Lab3	27	28
1 Hw5-due	2 Review for Exam3	3 Exam3 Lab3 report due		

ECE 2049 Syllabus

Class Schedule

<i>Class</i>	<i>Lecture Topics</i>
1	Introduction to Embedded Systems & Programming in C
2	Numbers
3	Endians
4	Architecture of MSP430 – CPU, memory, I/O
5	Data storage, and memory mapping
6	Basic Digital I/O
7	More Digital I/O
8	Making it work – Digital I/O
9	CPU Clock signals, clock set-up and the passage of time
10	Timers and Interrupts
11	More Timers & Interrupts
12	Intro to Data Acquisition Systems – ADC concepts
13	Using the on-chip A/D Converter
14	Interfacing ADC to devices (e.g. sensors, etc)
15	Event driven data acquisition
16	Introduction to serial interfaces, asynchronous serial communications
17	High speed synchronous serial interfaces (SPI/I ² C)
18	Operating Modes and the Watchdog Timer
19	Implementing a scheduled program
20	System Reset and initialization, operating modes